

2. Task 2 - Data Quality Assessment

2.1 Objective: To systematically audit data quality issues across missing values, duplicates, typos, and outliers.

2.2 Observation: Audited raw dataset and identified data quality issues summarized below.

Table 2. Summary of Data Quality Issues

Problem			
- Categorical fields contained multiple string variants (e.g. Mumbai vs Mumbai, Hyd vs Hyderabad).			

Table 3. IQR Outlier Summary

Attribute Column			

2.3 Key Inconsistencies & Domain Anomalies

3. Task 3 - Data Cleaning

3.1 Objective: To identify and correct missing values, duplicates, typos, noise, and outliers.

3.2 Observation: Implemented data cleaning pipeline. Table 5 records before/after comparison and reasons.

Table 5. Data-Cleaning Steps: Before, After, Reason, and Operations

Cleaning Step	Cleaned dataset exported successfully to 'cleaned_mall_customer_dataset.csv'.		

Table 5.1 Sample Cleaned Dataset Preview - First 5 Unique Rows

CustomerID							

3.4 Final Data Cleaning Validation Summary

4. Task 4 - Data Transformation

4.1 Objective: To convert cleaned data into machine-learning-ready format via encoding and feature scaling.

4.2 Observation: Applied Label Encoding, One-Hot Encoding (drop_first=True), and Min-Max Normalization.

Table 7. Feature Transformation Comparison

Transformation			

5. Task 5 - Data Reduction

Table 8. PCA Before-and-After Comparison

Measure		

6. Task 6 - Proximity Measures

7. Task 7 - Reflection Report
